

The RefinedWeb Dataset for Falcon LLM: Outperforming Curated Corpora with Web Data, and Web Data Only

G. Penedo¹, Q. Malartic², D. Hesslow¹, R. Cojocaru², A. Cappelli¹, H. Alobeidli², B. Pannier¹, E. Almazrouei², and J. Launay^{1,3}, ‘The RefinedWeb Dataset for Falcon LLM: Outperforming Curated Corpora with Web Data, and Web Data Only’, Jun. 01, 2023, arXiv: arXiv:2306.01116. doi: 10.48550/arXiv.2306.01116.

¹ LightOn

² Technology Innovation Institute, 9639 Masdar City, Abu Dhabi, United Arab Emirates

³ LPENS, École normale supérieure

Abstract

- LLMs are traditionally trained on a mixture of filtered web data and curated “high-quality” corpora
 - Curated sources include: social media conversations, books, technical papers
 - Curation process is believed to be necessary for performant models with broad zero-shot generalization
- Scalability concern: As models grow larger and require pre-training on trillions of tokens, it is unclear whether curation is scalable or if high-quality data will be exhausted
- Core hypothesis of this paper: Properly filtered and deduplicated web data alone can lead to powerful models
 - Models trained exclusively on filtered web data significantly outperform state-of-the-art models trained on The Pile
- Despite extensive filtering, high-quality data extracted from web remains plentiful
 - 5 trillion tokens successfully extracted from CommonCrawl
 - Availability does not appear to be a bottleneck for scaling

- Dataset Release: RefinedWeb
 - 600 billion tokens publicly released
 - 1.3B and 7.5B parameter language models trained on RefinedWeb released for community use
- Central claim: Web-only approach with proper filtering and deduplication is both scalable and superior to multi-source curation

[falcon-refinedweb](#)

| Introduction

Scaling Laws and Data Requirements

- NLP progress driven by compute scale; joint scaling of model and dataset size required
- Scaling challenge: optimal training of 175B parameter models requires ~3,500B tokens - far exceeding largest public datasets (OSCAR, C4, The Pile)

The Curation Dilemma

- Current practice: mix web crawls with “high-quality” curated sources (books, technical docs, social media)
 - Curation delivers quality but is labour-intensive, limited in scale, and raises licensing issues
- Conventional wisdom: web data is inherently lower quality than curated sources, even when filtered (C4, OSCAR)
 - Results in inferior model performance

Research Gap and Solution

- Challenge: sustain scaling needs while reducing human-intensive curation
- RefinedWeb premise: aggressive filtering and processing of web data alone can match or exceed performance of curated corpora
- This challenges fundamental assumptions about data quality hierarchies

Contributions

- RefinedWeb: 5 trillion tokens of filtered web-only English data; public release of 600B token extract + 1B/7B parameter models
- Empirical demonstration: web-only models outperform both public and private curated corpora on zero-shot benchmarks
- New baseline high-quality dataset for NLP community

Related Work

Pretraining Data Evolution

- Evolved from sentence-wise datasets → document-focused corpora (Wikipedia, BookCorpus) → massive web-scrape
- State-of-the-art: mixed aggregated corpora combining web data with curated sources (books, technical articles, social media); high-quality sources upsampled 1-5x
- Conventional belief: diversity and quality from curation essential; web-only data deemed insufficient

Web Data Filtering Pipelines

- Multi-stage filtering: language identification → heuristic rules → ML-based quality filtering → deduplication
- Result: reduces spam/pornographic content; improves model performance
- Caveat: excessive filtering introduces bias, may harm minority representation

Deduplication Strategies

- Methods: exact matching (suffix arrays), fuzzy matching (MinHash, SimHash), semantic embeddings
- Impact: reduces memorization; quality gains scale with model size (minor duplicates harmful at 175B parameters)
- Debate: limited evidence for deduplication benefit on curated corpora vs. web-based datasets

RefinedWeb Contribution

- Aggregates best-practices across existing pipelines; introduces line-wise corrections

- Large-scale combined exact + fuzzy deduplication (5 trillion tokens)
- Challenges assumption that web data is inferior to curated corpora

Macrodata Refinement and RefinedWeb

Overview

- MDR (MacroData Refinement): pipeline for filtering and deduplicating web data from CommonCrawl at scale
- RefinedWeb: 5 trillion tokens English pretraining dataset, web-only, matching quality of state-of-the-art curated corpora

Design Principles

- **Scale first:** target 3-6 trillion tokens for 40-200B parameter models; use CommonCrawl only, eschew labour-intensive human curation
- **Strict deduplication:** combine exact and fuzzy deduplication with high removal rates; inspired by Lee et al. (2022) showing deduplication's critical role for large models
- **Neutral filtering:** avoid ML-based filtering bias; use only simple rules/heuristics and URL filtering for adult content to prevent introducing undesirable biases

Document Preparation: Reading Data, Filtering URLs, Extracting Text, and Language Identification

Reading the Data

- Started with raw WARC files instead of preprocessed WET files
- Reasoning: WET files contain undesirable navigation menus, ads, irrelevant text despite preprocessing

URL Filtering

- Two-stage filtering: blocklist of 4.6M domains + URL score based on curated word lists (weighted by severity)
- Result: removes fraudulent/adult content while avoiding false positives (medical, legal, pop culture)

- Also removed common high-quality sources (Wikipedia, arXiv) to prevent overlap in aggregate datasets

Text Extraction

- Tool: trafilatura library (outperforms commercial alternatives for blog/news extraction)
- Processing: applied regex formatting to limit newlines and remove URLs

Language Identification

- fastText classifier (CCNet) at document-level; threshold: score > 0.65
- Result: RW-RAW stage—48% of original documents retained; majority filtered by language identification

Filtering: Document-wise and Line-wise

Repetition Removal

- Remove documents with excessive line, paragraph, or n-gram repetitions; implement heuristics from Rae et al. (2021)
- Goal: catch pathological repetition-induced behavior before deduplication stage

Document-wise Filtering

- Target: machine-generated spam, keyword lists, boilerplate text, special character sequences
- Method: apply quality heuristics (Rae et al., 2021) filtering outliers in length, symbol-to-word ratio, ensuring natural language
- Note: filters are language-dependent; naive transfer to other languages risks overfiltering

Line-wise Corrections

- Remove undesirable lines (social media counters, navigation buttons) not caught by trafilatura
- Threshold: if corrections remove >5% of document, discard entire document
- Result: RW-FILTERED stage

Data Retention

- RW-FILTERED: 23% of original CommonCrawl documents retained
- 50% of RW-Raw removed by filtering heuristics

| Deduplication: Fuzzy, Exact, and Across Dumps

Motivation & Challenge

- Duplicate content across documents favors memorization over generalization
 - Sources: crawler redundancy, boilerplate content (licenses), plagiarism
- Deduplication is computationally expensive with limited adoption in public datasets
- Strategy: aggressive combined approach integrating fuzzy and exact sequence removal

Fuzzy Deduplication (MinHash)

- Methodology: sketch-based similarity measurement across documents
 - Creates compressed document signatures for efficient similarity detection
 - Effectiveness: identifies templated documents
 - Licenses with varying entities
 - Placeholder SEO text repeated across websites
- Configuration: 9,000 hashes per document
 - 5-gram based calculations
 - 20 buckets of 450 hashes each
 - Note: parameter selection rationale documented in Appendix G.3.1
- Comparative findings: aggressive settings substantially outperform less aggressive variants
 - 10 hashes (The Pile baseline) result in lower deduplication rates
 - Performance degradation in downstream model evaluation
 - Aggressive configuration improves zero-shot performance

Exact Deduplication (Suffix Array)

- Scope: sequence-level token-by-token matching (vs. document-level)
 - Application: identifies exact substring matches
 - Specific disclaimers

- Standard notices
- Threshold: remove sequences >50 consecutive tokens
- Execution strategy: document modification via substring removal
 - Alternatives tested: full document dropping, loss-masking of duplicated strings
 - Result: negligible differences in zero-shot performance across approaches
 - Substring removal selected for implementation

URL Deduplication Across Dumps

- Implementation constraint: split CommonCrawl into 100 parts for computational feasibility
 - Stratified partitioning: each part contains 1/100th of each dump
 - Larger duplicate clusters (licenses, spams) distributed across parts, effectively removed through within-part deduplication
- Finding: significant cross-dump URL overlap despite identical content
 - URLs revisited across dumps without content changes
- Procedure: maintain URL registry from each processed part
 - Remove duplicate URLs from subsequently processed parts
 - Ensures cross-dump deduplication despite partitioned processing

Experiments

Evaluation Methodology

- Focus: zero-shot generalization across diverse tasks (vs. validation loss/perplexity alone)
 - Rationale: perplexity-optimization diverges from end-task performance
 - Perplexity measures token-level prediction accuracy (narrow, local signal)
 - End-task performance reflects practical capabilities (question answering, reasoning, code generation)
 - Models can achieve low perplexity while underperforming on downstream tasks
 - Alignment: matches modern LLM evaluation standard
 - Established by GPT-3, PaLM, BLOOM precedent (Brown et al., 2020; Rae et al., 2021; Chowdhery et al., 2022)
 - Shift from pre-GPT-3 era focus on validation perplexity
 - Reflects practical deployment metrics

- Natural setting: decoder-only models perform best in zero-shot regime (Wang et al., 2022)
- Infrastructure: Eleuther AI evaluation harness (Gao et al., 2021) enabling standardized zero-shot task coverage

Task Aggregates

Four task sets designed for different evaluation purposes:

Small Aggregate

- Purpose: internal ablation studies at small scale
- Selection criteria: tasks yielding non-zero performance at 1B parameters on 30GT training
- Rationale: ensures stable signal across filtering stages (raw → filtered → final)
- Use case: rapid ablation measurements (e.g., “Does deduplication improve performance?”)

Core Aggregate

- Purpose: broad model comparisons across contemporary models
- Basis: tasks commonly reported in modern model benchmarks (Dey et al., 2023; Biderman et al., 2023)
- Coverage: diverse task categories (language understanding, knowledge, reasoning)
- Use case: apple-to-apples comparison with OPT, BLOOM, Pythia, other public models

Main Aggregate

- Purpose: direct comparison with foundational LLM benchmarks
- Basis: tasks from GPT-3 and PaLM papers (Brown et al., 2020; Chowdhery et al., 2022)
- Significance: historical standardized evaluation protocol
- Use case: continuity with most influential prior work

Ext Aggregate

- Purpose: extended coverage from BigScience Architecture & Scaling research
- Basis: tasks established by Scao et al. (2022b) BLOOM development standards
- Scope: additional edge cases and domain-specific evaluations
- Use case: supplementary validation and robustness across task diversity

Comparison Levels & Methodological Rigor

- Level 1 – Internal comparisons (most controlled)
 - Scope: models trained and evaluated within same codebase
 - Variable: pretraining dataset only (RefinedWeb vs. The Pile)
 - Control: identical hardware, hyperparameters, training infrastructure
 - Validation: The Pile baseline models perform in-line with published results
 - Use case: Small aggregate ablation studies, cleanest dataset-level conclusions
- Level 2 – Benchmark-level comparisons (moderate control)
 - Scope: different training codebases, unified Eleuther AI harness evaluation
 - Sources: Scao et al. (2022b), Black et al. (2022), Aleph Alpha (2023), Dey et al. (2023)
 - Flag: * indicates Eleuther AI Harness evaluation
 - Caveat: training infrastructure differences may introduce externalities
 - Use case: Core and Main aggregates for broader model ecosystem comparison
- Level 3 – External comparisons (least controlled)
 - Scope: published results from independent papers
 - Sources: Brown et al. (2020), Chowdhery et al. (2022)
 - Flag: † indicates arbitrary evaluation setup (not standardized)
 - Caveat: numerical precision, prompt formulation, inference implementations may diverge
 - Use case: historical benchmarking and context, not primary evidence for claims

Model Architecture & Training Configuration

- Architecture: 1B, 3B, 7B parameter autoregressive decoder-only models
 - Base: GPT-3 hyperparameter configuration (Brown et al., 2020)
 - Position embeddings: ALiBi (Press et al., 2021)
 - Attention mechanism: FlashAttention (Dao et al., 2022)
 - Implementation: custom codebase for controlled comparison
- Baseline validation: The Pile models trained in custom codebase perform equivalently to published baselines
 - Ensures downstream results reflect dataset differences, not training infrastructure effects
- Training scale progression:
 - Ablation studies (1B, 3B models): 27B and 60B tokens respectively
 - Methodology: trained to optimality per Hoffmann et al. (2022) scaling laws
 - Small aggregate evaluation

- Main experiments (Falcon-RW models): 350GT tokens
 - Alignment: comparable compute budget to GPT-3, GPT-J, BLOOM public models
 - Aggregates: Core, Main, Ext evaluation
- Excluded comparisons: LLaMA models
 - Rationale: minimum LLaMA variant uses 2.5x larger compute budget
 - Impact: prevents meaningful dataset-level comparison

| Can Web Data Alone Outperform Curated Corpora?

Research Question

- Objective: demonstrate that web data alone can result in models outperforming other models trained on curated corpora

Methodology

- Phase 1: small-scale study with 1B and 3B parameter models trained to optimality (27B-Token and 60B-Token)
- Phase 2: scale to 1B and 7B-Parameter models trained on 350B-Token
- Comparison: zero-shot generalization against state-of-the-art models

Small-Scale Study

Experimental Setup

- Datasets tested:
 - Web-based public: OSCAR-2019, OSCAR-2022, C4
 - Curated: The Pile
 - RefinedWeb variants: RW-Raw, RW-Filtered, RW
- Control: all models trained with same architecture, same internal codebase, same evaluation framework
- Variable: pretraining dataset only
- Evaluation: results averaged on small aggregate (6 tasks), presented in Table 4

Findings

- Web datasets show relatively strong performance compared to The Pile
 - Observation: curation is not a silver bullet for performant language models
- C4: strong pretraining dataset, in line with BigScience findings
- The Pile: comparatively underperforms more in these benchmarks
- OSCAR-2022: relatively disappointing results
 - Possible cause: main version distributed without deduplication
- RefinedWeb: both filtering and deduplication significantly improve performance
 - Note: how much each component contributes measured separately in Section 4.3 (Ablation Studies)

Full-Scale Model Experiments

Experimental Setup

- Training: 1B and 7B-Parameter models on 350B-Token
- Control experiment: 1B-Parameter model on The Pile at 350B-Token budget
 - Purpose: test whether performance gains come from better training infrastructure or from better dataset quality

Baseline Models Compared

- GPT-3 series
- FairSeq series
- GPT-Neo/J series
- OPT series
- BigScience BLOOM
- PaLM-8B
- Aleph Alpha Luminous 13B
- Pythia series
- Cerebras-GPT series

Evaluation Methodology

- GPT-3 via API (Babbage, Curie models):
 - Babbage & Curie: smaller GPT-3 model variants (not full davinci)
 - Evaluated with EleutherAI harness: flagged *

- Paper-reported results: flagged †
- PaLM & OPT: results from paper with different evaluation setup: flagged †
- Other models: evaluated with EleutherAI harness: flagged *
- Note: * indicates standardized harness evaluation (more directly comparable); † indicates paper-specific setup

Results Presentation

- Main aggregate: Figure 1
- Core and extended aggregates: Figure 3

Key Findings

Performance comparison: - Open-source models consistently underperform private-curated models (e.g., GPT-3) - Observed even with standardized evaluation setup (* comparisons)

RefinedWeb vs. GPT-3: - RefinedWeb-trained models match GPT-3 series performance - Achieved using web data alone (no proprietary sources) - Despite excluding common high-quality sources in The Pile (Appendix Table 14)

Infrastructure validation (critical for interpretation): - Internal The Pile model trained in custom codebase: - Performs in line with BigScience BLOOM model - Interpretation: custom training infrastructure NOT superior - Why this matters: proves performance gains are NOT from better code - Conclusion: RefinedWeb performance improvements attributable to dataset quality itself - Primary drivers: filtering (removes low-quality content from Section 3.2) - Secondary driver: deduplication (removes duplicate content from Section 3.3) - Individual quantitative contributions: measured in Section 4.3 (Ablation Studies)

Figure 1. Models trained on R EFINE DW EB alone outperform models trained on curated corpora. Zero-shot performance on our main-agg task aggregate (see Section 4.1 for details). At equivalent compute budgets, our models significantly outperform publicly available models trained on ▼ The Pile, and match the performance of the ■ GPT-3 models when tested within our evaluation setup.

Figure 1. Models trained on R EFINE DW EB alone outperform models trained on curated corpora. Zero-shot performance on our main-agg task aggregate (see Section 4.1 for details). At equivalent compute budgets, our models significantly

outperform publicly available models trained on ▼ The Pile, and match the performance of the ■ GPT-3 models when tested within our evaluation setup.

Do Other Corpora Benefit from MDR?

Motivation & Challenge

- Research question: does MDR pipeline generalize to other popular pretraining datasets?
- Challenge: ablating MDR contributions difficult
 - Most heuristics lack agreed-upon ground truth
 - Individual component changes may be too small for sufficient zero-shot signal in pretraining
- Approach: apply filtering and deduplication stages independently to popular datasets
- Evaluation: small aggregate, results in Table 5

Overall Findings

- Filtering: not systematic improvements across datasets
- Deduplication: consistent positive effects across all datasets
- Combined effect: further improvements, though dataset-specific differences remain

Filtering Analysis

Performance Impact

- The Pile: improvements achieved
 - Adaptation necessary: line length and character ratio heuristics adjusted
 - Reason: prevent removal of books and code
- OSCAR-21.09: improvements
- C4: improvements
- OSCAR-22.01: performance worsens
 - Observation: removal rates do not strongly correlate with downstream accuracy

Interpretation

- Filtering effects inconsistent across datasets
- Heuristic generalization limited; dataset-specific tuning required

Deduplication Analysis

Performance Impact

- Steady boost across all datasets tested
- Removal rates better correlated with downstream accuracy changes

Duplicate Analysis by Dataset

- OSCAR-21.09: already well deduplicated
- C4: already well deduplicated
- The Pile: 40-60% duplicates
 - Consistent with Zhang et al. (2022) findings
- OSCAR-22.01 (base version): 40-60% duplicates
 - Distributed without deduplication

Interpretation

- Deduplication: more reliable improvement signal than filtering
- Significant duplicate prevalence in The Pile and OSCAR-22.01 explains gains

Combined Effect (Filtering + Deduplication)

Results

- Further improvements observed
- Performance more uniform across datasets

Limitations

- Dataset-specific differences persist
 - Interpretation: original text extraction and processing flaws cannot be fully compensated by MDR alone
 - Suggests: upstream data collection quality imposes lower bound on downstream performance

Case Study: C4 + MDR Processing

Combination Strategy

- C4 base filtering (stringent existing filters: NSFW word blocklist, 3-sentence span deduplication)
- Plus: MDR own filtering and deduplication

Results

- Subsets potentially outperform RefinedWeb slightly

Trade-off

- Combined approach: rejection rates too high for 3-6 trillion token target
- Implication: quality-quantity trade-off
- Alternative use case: shorter training runs where extremely high-quality data subsets can be extracted from large web datasets

Limitations

Biases

- Toxicity assessment of RefinedWeb via Perspective API (Figure 4)
 - Definition used: “content that is rude or disrespectful”
 - Finding: RefinedWeb toxicity comparable to The Pile
 - Scope limitation: analysis does not cover social biases or broader harmfulness
 - Note: detailed assessment methodology not specified in main text (see Figure 4)
- Assumption: MDR pipeline unlikely introduces additional issues beyond documented problems in popular datasets
- Recommendation: encourage further quantitative work on public RefinedWeb extract

Multiple Epochs

- Context: alternative source of training tokens – repeat same data across multiple passes (epochs) instead of acquiring unique tokens
 - What is an epoch: full pass through entire dataset
 - Example: Epoch 1 = model sees all data; Epoch 2 = model sees same data again

- Precedent: OPT, NeoX-20B train on up to 2 epochs; curated datasets typically upsample 2-5 times
- Trade-off with repetition: repeated exposure favors memorization over generalization
- Finding: Hernandez et al. (2022) shows 100B+ parameter models sensitive to even small number of epochs
- Open research questions:
 - Does aggressive deduplication (RW loses 77% of raw data) enable higher-quality multi-epoch training?
 - Is multiple epochs on higher-quality deduplicated data preferable to single epoch on lower-quality data?
- Further discussion: Appendix E.3

Other Deduplication Results

- Conflicting evidence: Biderman et al. (2023) found limited zero-shot performance impact from deduplicating The Pile
- Research gaps identified:
 - Further deduplication research needed on curated corpora
 - Effects of deduplication in data-constrained regime poorly understood (when multiple epochs necessary to compensate for token reduction)
- Further discussion: Appendix F.2

Conclusion

- LLM adoption driving demand for models trained beyond scaling law recommendations
 - Rationale: amortize inference costs through larger models
- Pretraining datasets with trillions of tokens increasingly essential
 - Target scale: order of magnitude beyond currently available public corpora
- Key demonstration: stringent filtering and deduplication produce five trillion token web-only dataset
 - Outcome: models competitive with state-of-the-art
 - Performance comparable to LLMs trained on curated corpora
- Public release: 600B-Token extract of RefinedWeb

- Real-world validation: RefinedWeb used to train state-of-the-art models
 - Example: Falcon-40B (Almazrouei et al., 2023) ## Discussion and Open Questions

Referenced Datasets

Dataset	Source/Origin		Purpose/Context in Paper	Token Count / Size
CommonCrawl	CommonCrawl Foundation		Primary raw source for RefinedWeb; 5 trillion tokens extracted	5T tokens extracted
The Pile	EleutherAI (2020)		Comparison baseline; RefinedWeb models outperform Pile-trained models	~825B tokens
RefinedWeb	TII (2023)		Main contribution; filtered and deduplicated web data	600B+ tokens (public release)
C4	Google (2019)	Research	Comparison baseline for filtering quality and model performance	~750B tokens